

Name: _____
 Class: _____
 Date: _____

Mr. Rawson • IB DP Physics

SHM 3 – States, Signs and Graphs

Answer Key

The last set was mostly about putting numbers into formulas. This one is about knowing where the oscillator *is* and what its graphs look like, which is where a large share of the marks in this topic actually sit. Show your working, and give every answer a unit.

1) **State** A mass on a spring oscillates vertically. Take **upwards as positive**. Complete the table with the sign of each quantity, using +, – or 0.

[4]

Where the mass is	x	v	a
at its lowest point	–	0	+
moving upwards through the equilibrium position	0	+	0
at its highest point	+	0	–
moving downwards, halfway between the top and equilibrium	+	–	–

Mark scheme

One mark per correct row. The acceleration always points back towards equilibrium, so its sign is always the opposite of the displacement, which is $a = -\omega^2 x$ read off a diagram rather than out of a formula.

2) **Identify** Each pair below describes an oscillator at one instant. State which two are **impossible** for simple harmonic motion, and explain why in each case.

[3]

A displacement maximum, acceleration maximum C displacement maximum, velocity maximum

B displacement zero, velocity maximum D displacement zero, acceleration maximum

Mark scheme

C and D are impossible.

C: at maximum displacement the oscillator has momentarily stopped, so the speed there is zero, not maximum. Maximum speed happens at the equilibrium position.

D: $a = -\omega^2 x$, so if $x = 0$ then $a = 0$. Zero displacement and maximum acceleration cannot occur together.

A and B are both fine, and are in fact the same instant described twice: A is an extreme, B is the equilibrium position.

3) **Determine** An oscillator is released from rest at maximum displacement at $t = 0$. Its period is 1.2 s. Determine the time at which it first passes through the equilibrium position, the time at which it first reaches maximum displacement on the other side, and the number of times it passes through equilibrium during the first 3.0 s.

Mark scheme

First through equilibrium after a quarter of a period, $T/4 = \mathbf{0.30\ s}$. First at the far extreme after half a period, $T/2 = \mathbf{0.60\ s}$.

In 3.0 s it completes $3.0/1.2 = 2.5$ oscillations. Equilibrium is crossed *twice* per oscillation, at 0.30, 0.90, 1.50, 2.10 and 2.70 s, so **5 times**. Counting periods and multiplying by two is quicker than listing, but list them if you are unsure.

- 4) **Calculate** An object performs simple harmonic motion with a period of 0.50 s and an amplitude of 8.0 cm. Calculate its maximum acceleration, and its acceleration at the instant its displacement is 3.0 cm.

[4]

Mark scheme

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.50} = 12.6\ \text{rad s}^{-1}, \text{ so } \omega^2 = 158\ \text{s}^{-2}.$$

$$a_{\text{max}} = \omega^2 x_0 = 158 \times 0.080 = \mathbf{13\ m\ s^{-2}}.$$

At $x = 3.0\ \text{cm}$, $a = -\omega^2 x = -158 \times 0.030 = -4.7\ \text{m s}^{-2}$, the minus sign meaning it points back towards equilibrium. Converting the centimetres to metres is where this one is usually lost.

- 5) **Determine** For a certain oscillator, a graph of acceleration against displacement is a straight line through the origin with a gradient of $-45\ \text{s}^{-2}$. Determine the angular frequency, the frequency and the period of the oscillation.

[3]

Mark scheme

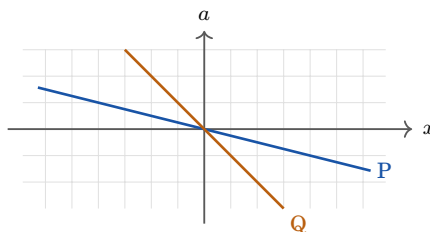
Comparing $a = -\omega^2 x$ with $a = (\text{gradient}) x$ gives $\omega^2 = 45\ \text{s}^{-2}$.

$$\omega = \mathbf{6.7\ rad\ s^{-1}}, \quad f = \frac{\omega}{2\pi} = \mathbf{1.1\ Hz}, \quad T = \frac{1}{f} = \mathbf{0.94\ s}.$$

Note that the amplitude never appears. An $a-x$ graph gives you ω and nothing else about how big the oscillation is.

- 6) **Sketch** On the axes below, sketch acceleration against displacement for two oscillators, **P** and **Q**, where Q has twice the angular frequency of P. Label each line. State what feature of a line tells you ω .

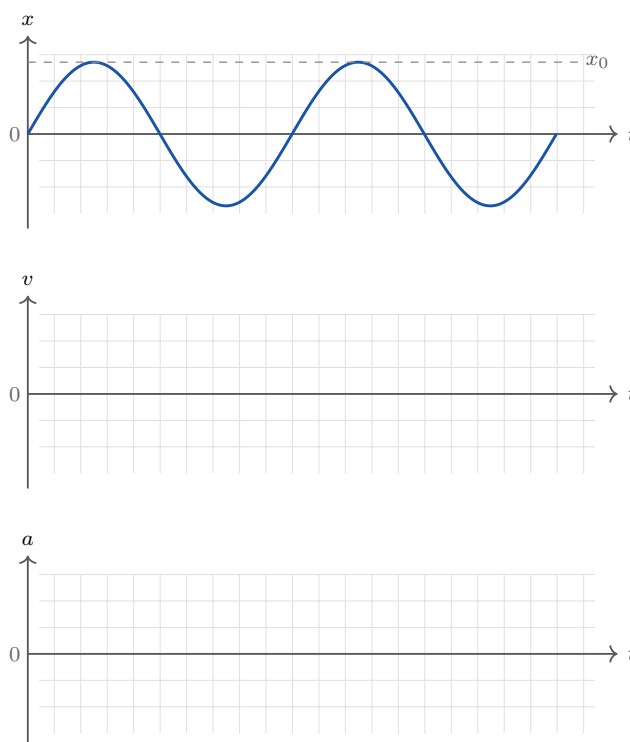
[3]

**Mark scheme**

Both are straight lines through the origin with *negative* gradient. The gradient is $-\omega^2$, so **Q's line is four times as steep as P's**, not twice, because the gradient goes as ω^2 . Steepness is what tells you ω ; where the line ends tells you the amplitude.

7) **Sketch** The displacement–time graph of an oscillator is given. Sketch its velocity–time and acceleration–time graphs on the axes beneath, keeping the same time scale. Then state the phase relationship between the displacement and the acceleration.

[5]



Mark scheme

The displacement is a sine starting at zero.

Velocity is the gradient of the displacement graph, so it is a *cosine*: maximum at every point where $x = 0$, and zero at every peak and trough. It leads the displacement by a quarter of a period.

Acceleration is the gradient of the velocity graph, so it is a negative sine: it is the displacement graph turned upside down and scaled by ω^2 .

Displacement and acceleration are therefore **exactly out of phase**, a phase difference of π radians, or half a period. That is the same statement as the minus sign in $a = -\omega^2 x$, drawn instead of written.

Higher level only

8) **Calculate** A mass of 0.20 kg hangs at rest on a spring of spring constant 40 N m^{-1} . It is pulled down a further 5.0 cm and released. Take $g = 9.81 \text{ m s}^{-2}$.

(a) Calculate the extension of the spring at the equilibrium position. (b) Calculate the extension at the lowest point of the oscillation. (c) Calculate the elastic potential energy stored in the spring at each of those two positions. (d) The mass loses 0.15 J of elastic potential energy on the way back up to equilibrium, but does not gain 0.15 J of kinetic energy. Explain why.

[7]

Mark scheme

(a) At rest the spring force balances the weight, so $ke = mg$ and $e = \frac{mg}{k} = \frac{0.20 \times 9.81}{40} = 0.049 \text{ m}$, which is **4.9 cm**.

(b) The lowest point is 5.0 cm below that, so the extension is $4.9 + 5.0 = \mathbf{9.9 \text{ cm}}$. *This is the step the question is really testing. The extension is measured from the natural length, the displacement is measured from equilibrium, and they differ by 4.9 cm at every instant.*

(c) $E_{\text{el}} = \frac{1}{2}ke^2$. At the lowest point, $\frac{1}{2} \times 40 \times 0.099^2 = \mathbf{0.20 \text{ J}}$. At equilibrium, $\frac{1}{2} \times 40 \times 0.049^2 = \mathbf{0.048 \text{ J}}$.

(d) Because the mass also *rises* by 5.0 cm, so some of the elastic energy released goes into gravitational potential energy, $mg\Delta h = 0.20 \times 9.81 \times 0.050 = 0.098 \text{ J}$. There are **three** energy stores here, elastic, gravitational and kinetic, and only their total is constant. An oscillator on a horizontal spring has two, which is why this vertical case catches people out.

Worth showing them that it closes exactly. The kinetic energy gained is $0.148 - 0.098 = 0.050 \text{ J}$. Independently, $\omega = \sqrt{k/m} = 14.1 \text{ rad s}^{-1}$, so $v_{\text{max}} = \omega x_0 = 0.71 \text{ m s}^{-1}$ and $\frac{1}{2}mv_{\text{max}}^2 = 0.050 \text{ J}$. The two routes agree to three decimal places, which is the proof that all three stores have been counted.